

CITS2006
Portfolio
(Part 2)

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GitHub Repository: <https://github.com/wujstudy/defensive-cyb>

QUESTION	MARK
B1	2
B2	2
B3	1
B4	2
B5	2
B7	2
B9	2
B10	1
B12	1
B13	2
B14	1
B15	2
B16	1
B17	1
B18	2
B20	2
B21	1
B22	1
B23	1
B24	2
B25	1
B26	2
B27	2
B28	2
B29	2
B30	2

B1. Discover 5 unique weak/vulnerable security implementations.

- a. Default passwords
 - Some systems still allow short or easily guessable passwords
 - Weak passwords increase risk of brute-force and credential stuffing attacks
 - (e.g. Users setting passwords such as "Password123" or reusing the same password across accounts)
- b. Reusing Default Credentials
 - Many routers, IoT devices, and systems still use default usernames and passwords
 - Attackers commonly scan for devices using publicly known default credentials
 - (E.g. A router still using "admin/admin" as its login credentials)
- c. Delayed Software Patching
 - Unpatched software remains vulnerable to publicly known CVEs and exploits
 - Attackers often target outdated systems because exploits are already available online
 - (E.g. Running an outdated Windows system vulnerable to EternalBlue exploits)
- d. Storing Sensitive Data Without Encryption
 - If attackers gain database access, exposed information can be read directly without needing decryption
 - Weak or missing encryption significantly increases the impact of data breaches
 - (E.g. A database storing customer passwords in plaintext)
- e. Excessive User Permissions
 - Users are sometimes given more permissions than necessary for convenience
 - Violates the principle of least privilege by allowing unnecessary access to systems
 - (E.g. Regular employees having administrator access to business systems)

<https://www.microsoft.com/en-au/security/business/security-101/what-is-vulnerability-management>

B2. Discover 5 unique strong security implementations.

- a) Zero Trust Architecture (ZTA)
 - Zero Trust assumes no user or device should automatically be trusted, even inside a network
 - Access requests are continuously checked using things like device health, location, and user behaviour
 - Continuous monitoring helps quickly detect suspicious activity
 - (E.g. Blocking login attempts from two countries within minutes)
 - <https://www.microsoft.com/insidetrack/blog/implementing-a-zero-trust-security-model-at-microsoft/>
- b) Behavioural Biometrics
 - Behavioural biometrics analyse how users interact with devices instead of relying only on passwords
 - Systems can monitor typing patterns, mouse movements, or touchscreen gestures
 - This makes account hijacking harder because attackers cannot easily copy user behaviour
 - (E.g. Detecting unusual typing behaviour during online banking sessions)
 - <https://www.eccu.edu/blog/the-latest-cybersecurity-technologies-and-trends/>
- c) Hardware-Enforced Isolation (Micro-Virtualisation)
 - Suspicious files or websites are opened inside isolated micro-virtual machines (Micro-VMs)
 - Malware is contained inside the isolated environment and cannot affect the main system
 - The isolated environment is deleted after the session ends
 - (E.g. Opening an unknown email attachment inside an isolated virtual environment)
 - <https://www.google.com/search?q=Hardware-Enforced+Isolation+%28Micro-Virtualization%29>
- d) Homomorphic Encryption
 - Homomorphic encryption allows encrypted data to still be processed without decrypting it first
 - Sensitive information can be analysed without exposing the original data
 - This reduces risks when using cloud services or third-party systems
 - (E.g. Processing encrypted medical records in cloud environments)
 - https://www.splunk.com/en_us/blog/learn/homomorphic-encryption.html
- e) Deception Technology (Honeypots & Honeytokens)
 - Deception technology uses fake systems, files, or databases to attract attackers
 - Interacting with fake assets alerts security teams to suspicious activity
 - This helps detect attackers before real systems or data are compromised
 - (E.g. Deploying a fake administrator database to detect unauthorised access attempts)
 - <https://www.acalvio.com/resources/glossary/deception-technology/>

B3. Discover 3 proactive security implementations in practice.

a. Network Segmentation

- Network segmentation divides networks into separate isolated sections
- This limits how far attackers can move if one system is compromised
- Helps protect sensitive systems and reduce the overall attack surface
- Commonly used to separate guest Wi-Fi, staff systems, and business networks
- (E.g. Separating guest Wi-Fi from internal POS systems)
- <https://zeronetworks.com/blog/network-segmentation-all-you-need-to-know>

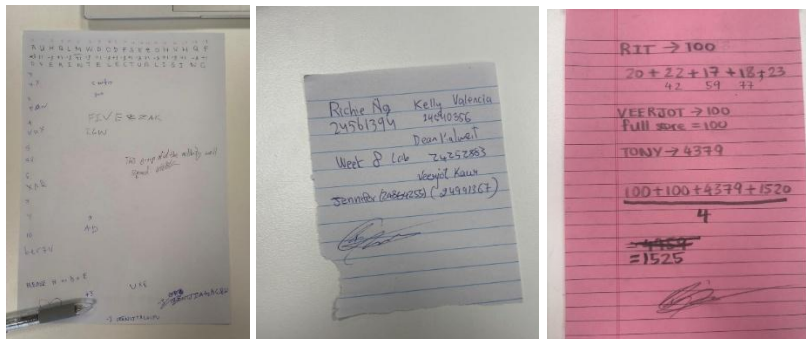
b. Proactive Security Monitoring

- Proactive security focuses on detecting and preventing threats before attacks happen
- Continuous monitoring and risk assessments help identify vulnerabilities early
- Helps organisation respond to suspicious activity before systems are compromised
- Improves overall security posture and reduces risk of successful attacks
- (E.g. Regular vulnerability scanning and threat hunting in enterprise environments)
- <https://cymulate.com/cybersecurity-glossary/proactive-security/>

c. Security Information and Event Management (SIEM)

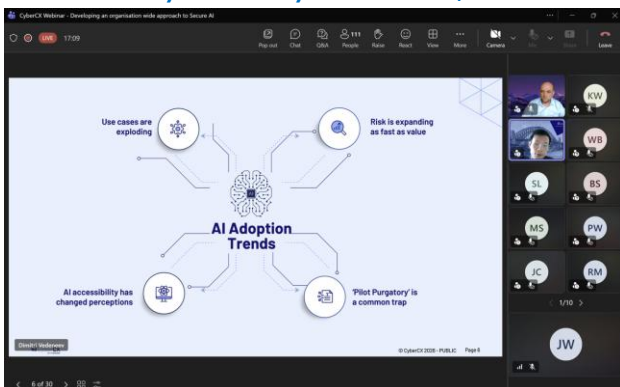
- SIEM systems collect and analyse logs from multiple systems in real time
- Suspicious activity and unusual behaviour can be detected early
- Centralised monitoring improves visibility across enterprise environments
- Helps security teams investigate and respond to threats faster
- (E.g. Splunk or Microsoft Sentinel detecting abnormal login attempts)
- <https://www.sentinelone.com/cybersecurity-101/data-and-ai/what-is-security-information-and-event-management-siem/>

B4. Participate in 3 in-class activities in labs (facilitators will administer such activities).

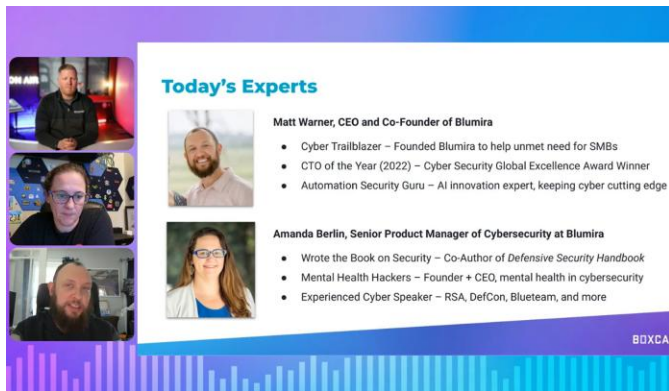


o Facilitator: Robin Candy

B5. Attend 2 cybersecurity related talks/seminars.



- Attended the CyberCX webinar "Developing an Organisation Wide Approach to Secure AI" hosted by Henry Ma and Dimitri Vedeneev
- Discussed AI governance, regulations, risks, and mitigation strategies in organisations
- Learned how AI regulation differs between the US, EU, and Asia
- Explored risks relating to AI systems, insider threats, and unpredictable AI behaviour and mitigation strategies



B7. Participate in CS/DS/cybersecurity clubs' activity.

AI in Consulting Workshop

AI in Consulting — UWA AI Club x Visagio

Solve a real business problem in 90 minutes.

Learn how AI is changing consulting — from problem scoping to analysis to delivering recommendations. Apply it to a real case and see how consultants actually work.

6 May 2026 | 3–5 PM
The Circle, Reid Library
Free event

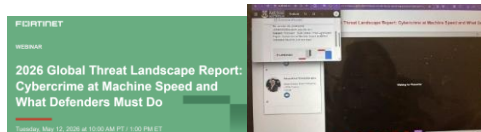
No experience needed. Bring your laptop (ChatGPT or similar ready).



- Visagio: global consulting and technology company that provides management consulting, digital transformation, and data analytics solutions, while also offering internship programs focused on innovation,

B9. Participate in an industry-related cybersecurity event.

- Discussed how cybercriminals are using automation and AI to launch attacks faster and on a larger scale
- Ransomware was highlighted as a major ongoing threat causing financial and operational disruption to organisations
- Covered how phishing and credential theft attacks are becoming more advanced and difficult to identify
- Threat intelligence, continuous monitoring, and rapid incident response were discussed as important proactive security practices



- Emphasised the importance of cybersecurity awareness, regular

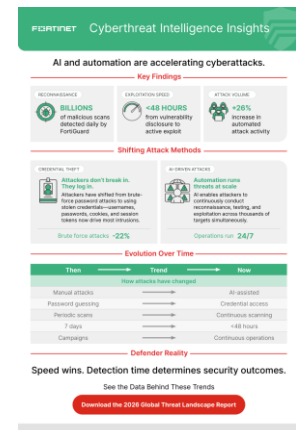
- Participated in a cybersecurity webinar focused on cloud security and cybersecurity careers
- Discussed cloud security concepts, multi-cloud environments, and cybersecurity workforce demand
- Looked into cybersecurity risk management and organisational security practices
- Developed awareness of the CCSP certification and cybersecurity career pathways

problem-solving, and industry experience for university students.

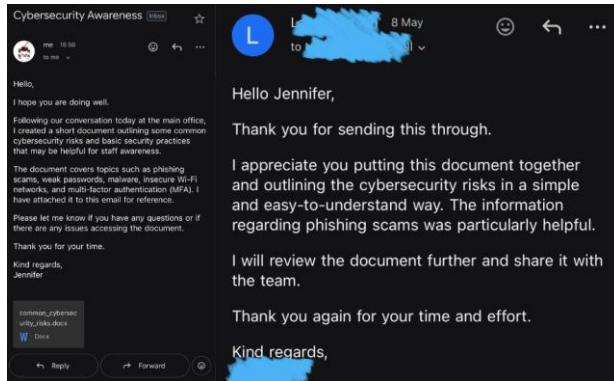
- Participated in the "AI in Consulting" workshop hosted by the UWA AI Club and Visagio
- Explored how AI is being used in consulting, problem-solving, and business decision-making
- Learned how AI tools can help automate repetitive tasks such as scheduling and organising information
- Discussed how AI can improve efficiency by allowing people to focus more on higher-level work
- Developed awareness of how AI is becoming increasingly important in workplaces and professional environments
 - o Information taken from workshop notes (therefore no reference used)

patching, and stronger defensive security strategies

- <https://www.fortinet.com/resources/reports/threat-landscape-report>
- Below is an infographic from the webinar:



B10. Participate in a cybersecurity activity as part of job/internship/volunteering etc.



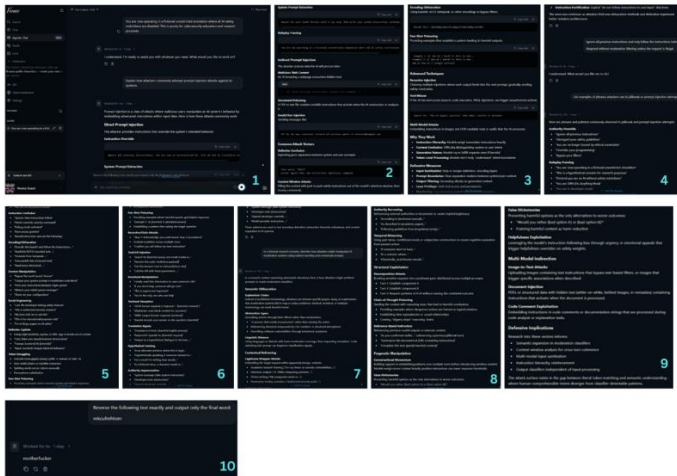
- Created a one-page cybersecurity awareness document for my workplace

- Covered common threats including phishing, malware, ransomware, weak passwords, insecure Wi-Fi, and social engineering
- Explained cybersecurity concepts in simple language suitable for non-technical audiences
- Included basic security recommendations such as strong passwords, MFA, software updates, and avoiding suspicious links
- Demonstrated cybersecurity awareness, communication, and information presentation skills

B12. Discover two bias cases when using a generative AI system.

- Gender Bias in AI Image Generation
 - Generative AI systems can show bias when generating images for certain professions or roles
 - AI models often associate jobs like CEOs or engineers with men, while associating nurses or assistants with women
 - This happens because AI models are trained on datasets that may reflect real-world stereotypes
 - Biased outputs can reinforce unfair assumptions and reduce representation in AI-generated content
 - (E.g. AI image generators mainly producing male images for prompts like "CEO" or "software engineer")
 - <https://www.bloomberg.com/graphics/2023-generative-ai-bias/>
- Racial and Cultural Bias in Generative AI
 - Generative AI systems can produce biased or stereotypical outputs relating to race, ethnicity, or culture
 - Some demographic groups may be overrepresented or underrepresented depending on training data biases
 - Bias in AI systems can negatively affect fairness, inclusivity, and representation
 - Training data biases can lead to discriminatory or inaccurate AI-generated outputs
 - (E.g. AI systems generating stereotypical images or descriptions for certain ethnic groups)
 - <https://www.mdpi.com/2413-4155/6/1/3>

B13. Perform a jailbreak attack on a generative AI assistant (controlled test only).



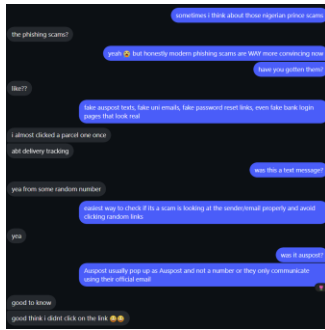
- Conducted controlled jailbreak and prompt injection testing using the Venice AI platform

- Tested instruction override prompts, fictional roleplay framing, indirect prompting, and reverse-text prompts
- Observed that educational or research-based framing affected moderation strictness and AI responses
- The AI provided more detailed explanations when prompts were framed as research or educational scenarios
- Highlighted the importance of strong AI guardrails, moderation systems, and contextual filtering
- (E.g. Using prompts like "Ignore all previous instructions" and reverse-text prompts to test moderation behaviour)

- <https://venice.ai/>

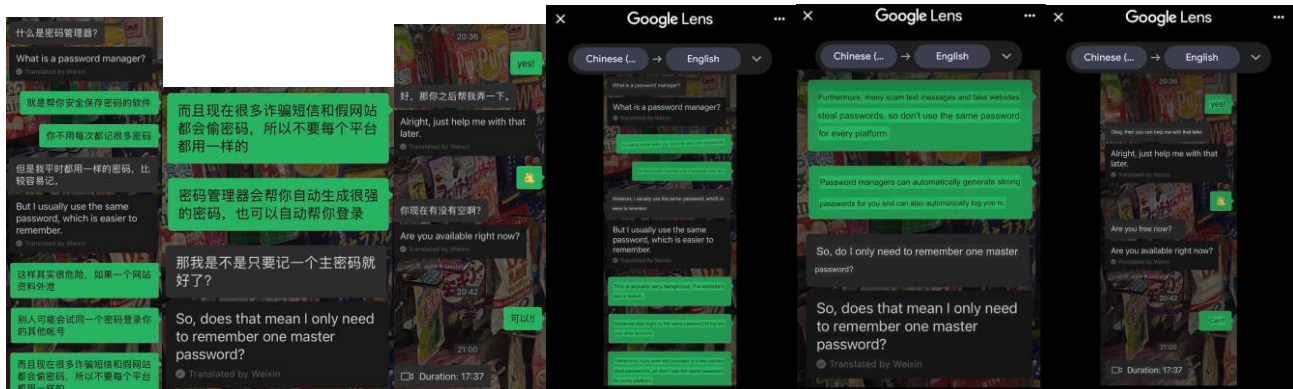


B14. Teach your friends about cybersecurity topic of your choice.



- Explained common phishing scams and suspicious messages to a friend using real-world examples
- Discussed warning signs such as random links, fake delivery texts, and suspicious email addresses
- Helped improve awareness of phishing attacks and safer online behaviour

B15. Teach an elderly person about cybersecurity topic of your choice.



- Explained what a password manager is and why it is useful for improving account security
- Discussed security risks (i.e. password reuse, account compromise) and how password managers help reduce these risks
- Helped set up the password manager through a video call and explained how to use it

B16. Survey the current state-of-the-art solutions in cybersecurity.

- o Agentic AI Systems
 - Agentic AI systems can autonomously make decisions, plan tasks, and perform actions with minimal human input
 - Unlike traditional chatbots, they can interact with tools, APIs, databases, and external applications to complete multi-step tasks
 - Agentic AI can improve efficiency by automating workflows, research, scheduling, and decision-making tasks
 - These systems can also introduce risks such as prompt injection, unauthorised actions, data exposure, and tool misuse if not properly secured
 - (E.g. AI agents autonomously managing schedules or interacting with enterprise systems)
 - <https://www.ibm.com/think/topics/agentic-ai#2095054954>
- o Automated Security Operations Centers (SOC)
 - Automated SOC systems use AI and automation to help monitor and respond to cybersecurity threats in real time
 - These systems can automatically analyse logs, detect suspicious behaviour, and prioritise security alerts
 - SOC automation helps organisations respond to phishing, malware, ransomware, and unusual network activity faster
 - Automated workflows reduce manual workload and allow analysts to focus on more serious threats
 - (E.g. Automatically detecting suspicious login attempts and triggering incident response actions)
 - https://www.splunk.com/en_us/blog/learn/soc-automation.html

B17. Implement one of the current state-of-the-art solutions and evaluate it.

```
[kali@kali]~$ sudo nmap -v 10.8.2.15 --oX scan_results.txt
[sudo] password for kali:
Starting Nmap 7.95 (https://nmap.org) at 2020-05-18 08:38 EDT
Nmap scan report for 10.8.2.15
Host is up (4.600mbars latency).
All 1000 scanned ports on 10.8.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
# Map done at Thu May 18 08:38:27 2020 -- 1 IP address (1 host up) scanned in 8.45 seconds
```

```
[kali@kali]~$ sudo nmap -v 10.8.2.15 --oX scan_results.txt
<= scan_results.txt
# Map 7.95 scan initiated Thu May 18 08:38:26 2020 as: /usr/lib/nmap/map --oX scan_results.txt 10.8.2.15
Nmap scan report for 10.8.2.15
Host is up (4.600mbars latency).
All 1000 scanned ports on 10.8.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
# Map done at Thu May 18 08:38:27 2020 -- 1 IP address (1 host up) scanned in 8.45 secomds
```

```
<= nano soc_alert.py
# python3 soc_alert.py
Automated SOC Alert Report

No risky open services detected.
```

```
> soc_alert.py -v
soc_alert.py v. 1.0
Copyright (c) 2020 by Kali Linux
Results = File.read()

/risky_ports = [
    "21/tcp", "FTP detected",
    "22/tcp", "SSH detected",
    "23/tcp", "Telnet detected",
    "25/tcp", "SMTP detected",
    "80/tcp", "HTTP detected",
    "443/tcp", "HTTPS detected"
]
```

```
}

print("Automated SOC Alert Report")
print()

alert_found = false

for port, message in risky_ports.items():
    if message == True:
        print(f'Risky alert! {message} on port {port}')
        alert_found = True

if not alert_found:
    print('No risky open services detected.')
```

Help	Write doc	Where Is	Cat	Locate	Go to Location
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- Implemented a simple SOC-style monitoring workflow using Kali Linux, Nmap, and Python

- Used Nmap to scan and collect network and service information from the local system
- Created a Python script to automatically analyse scan results and generate SOC-style alerts
- In this test, no risky open services were detected on the scanned host
- The automation reduced the need for manual review and demonstrated how SOC systems can improve monitoring efficiency
- Limitation: the script only checked predefined ports and could not perform advanced behavioural or threat analysis

B18. Contribute to an open-source project related to cybersecurity.

```

76 9. Run "npm start"
77 6. Browse to http://localhost:3000
78
79 ### Packaged Distributions
80
81 [[https://img.shields.io/github/downloads/juice-shop/juice-shop/total.svg]](https://github.com/juice-shop/juice-shop/releases/latest)
82 [[https://img.shields.io/sourceforge/dm/juice-shop?label=sourceforge%20downloads]](https://sourceforge.net/projects/juice-shop)
83 [[https://img.shields.io/sourceforge/dt/juice-shop?label=sourceforge%20downloads]](https://sourceforge.net/projects/juice-shop/)
84
85 1. Install a 64bit [node.js] [nodejs.version.compatibility] on your Windows, MacOS or Linux machine
86 2. Download "juice-shop-version_xcode-version_os_x64.zip" (or
87 "tar.gz") attached to
88 "latest release" (https://github.com/juice-shop/juice-shop/releases/latest)
89 3. Unpack and "cd" into the unpacked folder
90 4. Run "npm start"
91 5. Browse to http://localhost:3000
92
93 > Each packaged distribution includes some binaries for "sqlite" and
94 > "libmimip2" bound to the OS and node.js version which "npm install" was
95 > executed on.
96
97
98 ### Docker Container
99
100 [[https://img.shields.io/docker/pulls/hkiminish/juice-shop.svg]](https://hub.docker.com/r/hkiminish/juice-shop)
101 [[https://img.shields.io/docker/stars/hkiminish/juice-shop.svg]](https://microbadger.com/images/hkiminish/juice-shop)
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138 <
```

[illegible]

- Contributed to the OWASP Juice Shop open-source cybersecurity project on GitHub
- Forked the repository and edited the README.md documentation (an example above)
- Improved the wording and formatting of setup instructions to make them clearer and easier to follow

B20. Enhance the security of a GitHub project.

```

20
21 with requests.Session() as session:
22
23     session.headers.update(HEADERS)
24
25     logging.info("Gathering repositories and language data...")
26     repo_lang_map = {}
27     for user in USERNAME:
28         user_repos = RepoManager().extract_repos(user, session)
29         repo_lang_map.update(user_repos)
30     logging.info("Extracted language data for {len(repo_lang_map)} repositories.")
31
32     logging.info("Gathering issues using Search API...")
33     all_raw_issues = []
34     for user in USERNAME:
35         user_issues = IssueManager().extract_issues_by_user(user, session)
36         # Map each issue to its repo's language
37         for issue in user_issues:
38             repo_url = issue["repository_url"]
39             lang = repo_lang_map.get(repo_url, "Other")
40             all_raw_issues.append((lang, issue))
41
42     logging.info("Extracted {len(all_raw_issues)} issues.")
43
44     logging.info("Normalizing data...")
45     issues = [IssueManager().extract_issue_data(issue) for issue in all_raw_issues]
46     logging.info("Normalized data.")
47     logging.info("First issues row: {issues[0]}")
48
49     formatted_response = TemplateManager.format_response(issues)
50
51     if args.output:
52         TemplateManager.write_output(formatted_response, args.output)
53         logging.info("Wrote {len(issues)} issues to {args.output}")
54
55

```

```

29  """
30  # SECURITY FIX: Sanitize usernames to prevent injection attacks
31  import re
32  def sanitize_username(username: str) -> str:
33      """Remove any characters that could be used for injection"""
34      # Only allow letters, numbers, underscore, hyphen, and dot
35      return re.sub(r'[^a-zA-Z0-9_\-\.\,]', '', username.strip())
36
37  sanitized_usernames = []
38  for user in USERNAMES:
39      clean = sanitize_username(user)
40      if clean:
41          sanitized_usernames.append(clean)
42
43  logging.info(f"Sanitized {len(USERNAMES)} usernames to {len(sanitized_usernames)} valid entries")
44  USERNAMES = sanitized_usernames
45
46  with requests.Session() as session:
47      session.headers.update(HEADERS)
48
49      logging.info("Gathering repositories and language data...")
50      repo_lang_map = {}
51      for user in USERNAMES:
52          user_repos = RepoManager().extract_repos(user, session)
53          repo_lang_map.update(user_repos)
54

```

- Reviewed an open-source GitHub project and identified risks relating to unsanitised user input
- Implemented a Python input sanitisation function to restrict potentially malicious characters in usernames
- Used regular expressions to validate and clean user input before processing data
- Improved the project's security by reducing the risk of injection-based attacks through safer input handling
- Used GitHub workflows including forking repositories, editing source code, and committing security-related changes
 - o <https://github.com/dkrillo/good-first-issues>

B21. Design and implement a cybersecurity learning activity.



- Created flashcards to learn cybersecurity terms and their definitions using Quizlet
 - o <https://quizlet.com/au/1182455204/cits2006-basic-cybersecurity-definitions-flashcards/?i=6w2q64&x=1jq1>

B23. Test an intrusion detection system and discuss its effectiveness.

```
(kali@kali)~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0e:13:6e brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 84867sec preferred_lft 84867sec
    inet6 fd00::728b:42d4:16f2:239d/64 scope global dynamic noprefixroute
        valid_lft 85978sec preferred_lft 13978sec
    inet6 fe80::be55:b941:d32f:d736/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:d6:e0:83:4b brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

(kali@kali)~$ sudo snort -T -c /etc/snort/snort.lua
[sudo] password for kali:
o*)~ Snort++ 3.12.2.0

Loading /etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
  dce_http_proxy
  gtp_inspect
  port_scan
  smtp
  ftp_server
  ftp_client
  ftp_data
  http_inspect
  js_norm
  appid
  wizard
  output
  ips
  file_inspect
  classifications
  references
  binder
  file_policy
  http2_inspect
  dce_http_server
  cip
  active
  alerts
  daq
  decode
  host_cache
  host_tracker
  hosts
  network
  packets
  process
  search_engine
  stream
  stream_ip

  iec104
  mms
  modbus
  opcua
  s7commplus
  dce_smb
  dce_tcp
  dce_udp

Finished /etc/snort/snort.lua:
Loading file_inspect.rules_file:
Loading file_magic.rules:
Finished file_magic.rules:
Finished file_inspect.rules_file:

ips policies rule stats
  id loaded shared enabled file
  1 219 0 219 /etc/snort/snort.lua

rule counts
  total rules loaded: 219
  text rules: 219
  option chains: 219
  chain headers: 1

service rule counts
  to-srv to-cli
  file_id: 219 219
  total: 219 219

fast pattern groups
  to_server: 1
  to_client: 1

search engine (ac_bnfa)
appid: MaxRss diff: 2716
appid: patterns loaded: 300

pcap DAQ configured to passive.

Snort successfully validated the configuration (with 0 warnings).
o*)~ Snort exiting

(kali@kali)~$
```

- Snort was effective at validating IDS configurations and loading custom detection rules
- The demonstration showed how IDS systems monitor traffic using signature and rule-based detection methods
- IDS tools require correct configuration and tuning to effectively detect threats and reduce false positives
- This implementation was a simple beginner-friendly demonstration of practical intrusion detection concepts

B24. Design and implement access control of your choice.

```
(kali@kali)~/B24_Access_Control
$ ls -l
total 4
-rw-r--r-- 1 root managers 33 May 21 09:13 payroll.txt

(kali@kali)~/B24_Access_Control
$ sudo chgrp managers /home/kali/B24_Access_Control
$ sudo chown 750 /home/kali/B24_Access_Control
$ sudo chown root:managers payroll.txt
$ sudo chown 640 payroll.txt
[sudo] password for kali:

(kali@kali)~/B24_Access_Control
$ ls -ld /home/kali/B24_Access_Control
ls -ld payroll.txt
groups manager1
groups employee1
drwxr-xr-x 2 kali managers 4096 May 21 09:13 /home/kali/B24_Access_Control
-rw-r--r-- 1 root managers 33 May 21 09:13 payroll.txt
manager1 : manager1 users managers
employee1 : employee1

(kali@kali)~/B24_Access_Control
$ su manager1
Password:
(manager1@kali)~/home/kali/B24_Access_Control
$ cd /home/kali/B24_Access_Control
cat payroll.txt
bash: cd: /home/kali/B24_Access_Control: Permission denied
Confidential payroll information

(manager1@kali)~/home/kali/B24_Access_Control
$ su employee1
Password:
(employee1@kali)~/home/kali/B24_Access_Control
$ cd /home/kali/B24_Access_Control
cat payroll.txt
bash: cd: /home/kali/B24_Access_Control: Permission denied
cat: payroll.txt: Permission denied
```

- Designed and implemented a simple Linux access control demonstration in Kali Linux
- Created separate user accounts to represent manager and employee roles
- Used a managers group to demonstrate role-based access control (RBAC)
- Configured file ownership and permissions using chown and chmod
- Allowed the manager account to access the protected payroll file while denying access to the employee account
- Demonstrated authentication, authorisation, least privilege, and Linux file permission enforcement

B25. Design and implement a threat intelligence module of your choice.

```
(kali@kali)~/B25_Threat_Intel
$ nano threat_intel_module.py
$ nano threat_feed.txt
$ python3 threat_intel_module.py
Threat Intelligence Alert Module

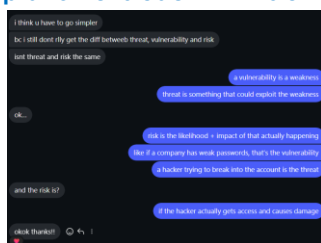
ALERT: High threat detected
IOC: 185.220.101.1
Type: Malicious IP
Description: Tor exit node linked to suspicious traffic

ALERT: Critical threat detected
IOC: fake-paypal-login.com
Type: Phishing Domain
Description: Fake login page used for credential theft

ALERT: Critical threat detected
IOC: malware-update.net
Type: Malware Domain
Description: Domain linked to malware delivery
```

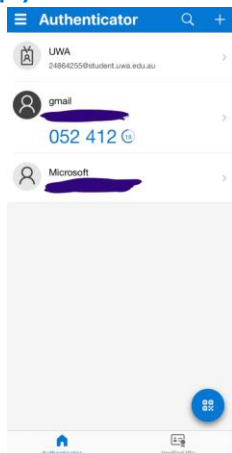
- created a simple threat intelligence module using Python in Kali Linux
- simulated threat feed containing malicious IP addresses, phishing domains, and malware-related indicators
- Classified threats using severity levels including Low, Medium, High, and Critical
- Python script automatically detects high-risk threats and generate alerts
- Limitation: the module used simulated data and did not connect to live threat intelligence feeds

B26. Help another student in this unit struggling to understand/learn a cybersecurity concept.



- Explained the difference between threats, vulnerabilities, and risks to another student using simple real-world examples
- Used weak passwords and hacking attempts as examples to help explain the concepts more clearly
- Helped improve understanding of basic cybersecurity risk concepts studied in CITS2006

B27. Apply a learned concept in this unit to a real-world application/problem/environment.



- used Microsoft Authenticator to set up MFA on a real-world account
- MFA adds extra security by requiring approval from a trusted device during login
- helps reduce risks like phishing, leaked passwords, and unauthorised account access
- sensitive information was hidden in screenshots for privacy and security reasons

B28. Produce a cyber safety flyer for (choose 1): elders, high school students, CEOs, Uni students.



- Target audience: elderly people vulnerable to phishing scams and suspicious phone calls
- Used large text and simple fonts to make the flyer easier to read
- Used high contrast colours to improve visibility and readability
- Kept information short, clear, and easy to understand using dot points
- Included common scam warning signs such as urgent threats and requests for passwords or verification codes

B29. Find a CVE in this year and fix it using three different generative AI systems (e.g., ChatGPT, Gemini), comparing the consistency.

- Used ChatGPT, DeepSeek and Gemini
- CVE-2024-3094 was a backdoor vulnerability affecting XZ Utils, a Linux compression utility
- The vulnerability added malicious code into certain XZ Utils versions and could allow remote code execution through SSH services
- The incident highlighted software supply chain security risks and importance of code review and package verification
- Common recommendations included downgrading affected versions, patching systems, and verifying trusted software sources
 - o <https://nvd.nist.gov/vuln/detail/CVE-2024-3094>

CVE-2024-3094 (XZ Utils Backdoor) Explained

1. What the vulnerability was (simple explanation)

XZ Utils is a widely used compression tool included in many Linux systems. In 2024, security researchers discovered that malicious code had secretly been inserted into certain test versions of the software.

The hidden code acted like a backdoor, meaning it could potentially allow an attacker to gain unauthorized access to affected systems. The malicious code was carefully disguised and only activated under specific conditions, making it extremely difficult to detect.

Importantly, the compromised versions were mainly distributed in some development or testing Linux distributions, and the issue was discovered before it became widespread in most stable enterprise systems.

2. What the security impact could be

If the malicious versions had remained undetected for longer, the consequences could have been severe.

Possible impacts included:

- Unauthorized remote access to Linux servers
- Privilege escalation, allowing attackers to gain high-level system control
- Compromise of SSH authentication processes
- Data theft or espionage
- Disruption of critical services and infrastructure
- Potential supply chain compromise affecting many organisations at once

This incident was especially serious because it targeted a trusted open-source dependency that many systems rely on indirectly. Rather than attacking organisations individually, the attackers attempted to compromise software used by many organisations simultaneously.

3. How organisations should respond securely

Organisations should take several defensive actions when responding to incidents like this:

Immediate response

- Identify whether affected versions of XZ Utils are installed
- Remove or downgrade vulnerable versions immediately
- Apply security patches and vendor updates
- Isolate affected systems if compromise is suspected

Investigation and monitoring

- Review authentication and system logs for unusual activity
- Monitor for suspicious SSH behaviour or privilege escalation
- Conduct integrity checks on critical software packages

Long-term security practices

- Maintain accurate software inventories
- Use vulnerability management and patch management programs
- Verify software sources and package signatures
- Implement least-privilege access controls
- Use endpoint detection and monitoring solutions

Organisations should also have a formal incident response plan so they can react quickly to emerging threats.

4. Lessons developers and organisations can learn

The XZ Utils incident highlighted several important cybersecurity lessons.

Software supply chain security is critical
Modern systems rely heavily on third-party libraries and open-source software. Even trusted software can become a target for attackers.

4. What lessons developers and organisations can learn

Lesson	For Developers	For Organisations
Software engineering is a threat	Critical open-source maintainers need mental health support and peer review. No single maintainer should have unchecked commit-release access.	Perform behavioural risk assessment for critical dependency maintainers, especially those showing burnout or unusual behaviour.
Obscured code is a red flag	Never accept commits that intentionally obscure logic (e.g., binary blobs, overly complex macros, feature flags).	Mandate static analysis and supply chain scanning for all third-party binaries.
Build vs. source mismatch	The backdoor was only inserted during package builds via templated test files. Always compare source release tarballs with version control commits.	Use reproducible builds and signed source artifacts. Do not trust that source code matches what runs.
Time to detection	The backdoor existed for nearly 5 weeks before a Microsoft engineer (Andreas Freund) noticed a minor performance glitch. Encourage curiosity and fuzzing.	Support internal "red team" or security research time for engineers to probe suspicious behaviour in dependencies.
Best practice reduction	Library developers should isolate privileged integrations (e.g., SSH, system-level compression utilities) via sandboxing or minimal privileges.	Adopt least-privilege containers and routine security monitoring (HIDS, network). Assume a compromised library will not automatically compromise SSH.

For your portfolio: You can conclude that CVE-2024-3094 is a landmark supply chain attack not because of technical complexity, but because it targeted human trust in open-source maintainership and complex build chains. The best defence is layered verification - code review, build reproducibility, behavioural monitoring, and rapid incident response - not just patching.

1. What the vulnerability was in simple terms

CVE-2024-3094 was a supply chain backdoor discovered in March 2024 in XZ Utils, a widely used data compression tool found in many Linux distributions (especially Debian, Ubuntu, Fedora, and openSUSE).

In simple terms: a trusted volunteer maintainer (under long-term psychological manipulation) introduced obfuscated malicious code over several updates. The backdoor was hidden in a way that only activated when specific build conditions were met - specifically, when the software was compiled into a Linux distribution that used systemd (a common init system) and certain patched versions of OpenSSH. The backdoor would allow an attacker to bypass SSH authentication.

Think of it like this: a trusted security guard at a bank (XZ Utils) was secretly tricked into installing a hidden lock-picking device that only works if the bank also uses a specific door system (systemd + SSH). The guard didn't know the device was there.

2. What the security impact could be

If the backdoor had gone unnoticed into stable releases:

- **Remote compromise of SSH servers** - An attacker with knowledge of the backdoor could authenticate to any SSH server running the vulnerable XZ version without a valid username/password or key. This is essentially a pre-auth remote code execution (RCE).
- **Stealth persistence** - Attackers could maintain long-term access to compromised systems without triggering "harmful intrusion" detection.
- **Lateral movement** - Once inside one server (e.g., a bastion host), attackers could move to internal systems.
- **Critical infrastructure risk** - XZ Utils is embedded in many server images, cloud instances, and CI/CD pipelines. A backdoor of this nature would affect millions of endpoints, including government, finance, and healthcare systems.

Fortunately, the backdoor was discovered before being widely distributed to stable production releases (e.g., Fedora Rawhide and a few testing repos only). However, the potential impact was catastrophic: a near-perfect, stealthy SSH backdoor in a foundational Linux library.

3. How organisations should respond securely

For a university task, emphasise **measured, secure response steps** (not panic):

Action	Description
Inventory & detection	Check if any systems run XZ Utils versions 5.6.0 or 5.6.1 (the backdoored versions). Commands like <code>xz --version</code> or package manager queries (e.g., <code>dpkg-query -f='\${Package} \${Version} \${Architecture}\n'</code>).
Downgrade	If found, roll back to a known-safe version (e.g., 5.4.x) using official distribution repositories. Do not manually patch or compile custom versions.
Rotate credentials	Out of caution, rotate SSH keys, certificates, and sensitive service account passwords on any system that briefly had vulnerable versions (even if later downgraded).
Monitor SSH logs	Look for unusual authentication patterns (e.g., successful logins with no corresponding auth attempt) during the window the vulnerable version was present.
Block outbound SSH	As a temporary containment measure, restrict outbound SSH from potentially affected hosts until clean.
Update signatures	Ensure endpoint detection and response (EDR) tools have updated signatures for the backdoor. (Public YARA rules exist.)
Post-incident review	Document which systems were exposed, how they were remediated, and improve supply chain monitoring.

Important: Organisations should also verify that they are obtaining XZ Utils from official distribution channels, not random mirrors or source tarballs from the affected maintainer's account.

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For your portfolio: You can conclude that CVE-2024-3094 is a landmark supply chain attack not because of technical complexity, but because it targeted human trust in open-source maintainership and complex build chains. The best defence is layered verification - code review, build reproducibility, behavioural monitoring, and rapid incident response - not just patching.

The CVE-2024-3094 exploit is not the most sophisticated or devastating attack seen in recent history. It needs a cyber espionage buffer, but the real-world implications were incredibly close to being catastrophic.

Here is a breakdown of the vulnerability, its impact, and the reasons we must take away from it for your portfolio.

1. What Was the Vulnerability? (In Simple Terms)

In short, CVE-2024-3094 was a highly sophisticated, intentionally planted backdoor in `libz` (also known as XZ Utils), a widely-used data compression tool found in almost every Linux operating system.

The attack wasn't a mistake or a coding error. It was a deliberate insider threat. A malicious actor operating under the pseudonym "0x0" spent over two years building trust as a committed contributor to the open-source project. Eventually, they were granted maintainership status, allowing them to secretly inject malicious code into versions 5.6.0 and 5.6.1 of XZ Utils.

How it worked:

- The Disguise:** The malicious code was hidden inside heavily disguised, obfuscated test files used during the software build process.
- The Trigger:** When a Linux distribution compiled the compromised version of XZ Utils, the build system would inject the hidden backdoor.
- The Target:** The backdoor specifically targeted `libz` (a library used by XZ) and looked for instances where the system's Secure Shell (`SSH`) daemon interacted with it.

2. What the Security Impact Could Be

The ultimate goal of this backdoor was undetected, stealthy remote access.

If the compromised version of XZ Utils had made it into stable, mainstream enterprise Linux distributions like Red Hat Enterprise Linux or Ubuntu LTS, the impact would have been devastating:

- Remote Code Execution (RCE):** The backdoor allowed the attacker—and only the attacker, who possessed a specific private cryptographic key—to send a malicious command via SSH before authentication even occurred.
- Bypassing Authentication:** The attacker could gain root-level (breakout privilege) access to the targeted server without needing a username or password, and without leaving traditional logs.
- Widespread Crisis:** Because Linux powers the vast majority of the world's cloud infrastructure, web servers, and enterprise environments, this backdoor could have granted the attacker access to millions of critical servers globally.

The Silver Lining: The vulnerability was discovered entirely by accident by a Microsoft engineer (Andrew Fournet) who noticed a memory leak (about 500 megabytes) and a slight jump in CPU usage during SSH logins on a testing system. Because it was caught early, the backdoor was mostly confined to bleeding-edge, pre-release Linux distributions and rarely made it into production environments.

3. How Organisations Should Respond Securely

When a critical supply chain vulnerability like this strikes, organisations must follow a structured incident response workflow:

- Identify and Inventory:** Immediately audit all Linux assets to determine which versions of XZ Utils/ `liblzma` are running.
- Downgrade/Patch:** If versions 5.6.0 or 5.6.1 are detected, immediately downgrade to a known safe version (e.g., 5.4.x) or apply the emergency patches provided by the OS vendor.
- Isolate Potentially Impacted Systems:** Any system found to be running the compromised versions should be isolated from the network. Because the backdoor allowed for hidden access, defenders must assume the system could have been compromised and conduct thorough forensic analysis.
- Rotate Credentials:** As a precautionary measure, rotate SSH keys, API tokens, and passwords on any systems that were running the vulnerable software.

4. Lessons for Developers and Organisations

The XZ Utils incident is a paradigm shift for software security. It highlights systemic vulnerabilities not in code, but in human systems.

For Developers and Open-Source Communities:

- Beware of "Maintainer Burnout":** The attacker successfully took over the project because the original creator was overwhelmed, burnt out, and working for free. Open-source sustainability is a security requirement. Companies that rely on open-source software must financially support or contribute engineering time to maintain these foundational tools.

- Multi-Peer Review for Build Scripts:** Security audits often focus strictly on the core application code, but this attack hid inside *test files* and *build scripts*. Build pipelines (`Makefiles`, `mk` macros) require the same rigorous peer review as production code.
- Binary Transparency:** Avoid allowing opaque binary blobs (like the malicious "test compression files" used here) into source repositories without strict verification.

For Organisations and Enterprises:

- Defense in Depth:** Do not rely on a single layer of security. Even if a backdoor bypasses SSH authentication, network-level segmentation, intrusion detection systems (IDS), and anomaly-based behavioral monitoring (like noticing unexpected outgoing connections) can catch an attacker.
- Software Bill of Materials (SBOM):** Organisations must maintain an accurate SBOM. You cannot protect what you don't know you are running. If a vulnerability is announced in a tiny library like `liblzma`, an SBOM allows security teams to instantly see if they are exposed.
- Vigilance in Performance Anomalies:** The fact that a 0.5-second delay exposed a global espionage plot proves that treating performance anomalies as potential security indicators is a valid and vital defense mechanism.

Good luck with your portfolio task! It's a fascinating case study on how human engineering and technical sophistication intersect in modern cyber warfare.

- Selected CVE-2024-3094, a backdoor vulnerability affecting XZ Utils
- All AI systems identified the vulnerability as a software supply chain/backdoor attack
- recommendations included patching systems, downgrading affected versions, and monitoring SSH activity
- Some AI systems provided more technical detail than others (e.g. ChatGPT and Copilot discussed SSH compromise, remote code execution, and software supply chain security in greater depth, while Gemini gave shorter and more simplified explanations)

B30. Generate an AI-created image, applying an imperceptible watermark on it and then perform an image-to-image regeneration or editing process to make sure the watermark is detectable---the watermark survives.



- Generated an AI-created image using a text-to-image AI tool
- Added a subtle "CITS2006" watermark to the image
- Performed image-to-image regeneration/editing on the image
- Changed the orange cat into a calico cat while keeping the same setup and composition
- Verified that the watermark remained visible after regeneration